Chpater4 Optics & Light

Lesson1 Light Propagation & Reflection

1. What is light, and how does it travel?
2. What are the sources of light, and give some examples.
3. Explain the difference between luminous sources (produce their own light) and non-luminous sources (depend on light from other sources).
4. A pinhole camera of length $20\ \text{cm}$ is used to view the image of a tree of height $12\ \text{m}$ which is $40\ \text{m}$ away from the pinhole. Calculate the height of the image of the tree obtained on the screen.
5. If the pinhole camera, in Example 1 is moved by $10\ \text{m}$ towards the tree, what will be the height of the tree on the screen?
6. A pinhole camera is used to take the photograph of a person who is $4\ \text{m}$ away from the pinhole. If the length of the box used is $18\ \text{cm}$ and the height of the image of the person is $9\ \text{cm}$, calculate the:
   (a) magnification produced by the pinhole camera. **(b)** height of the person.
7. A tree 18 m high is observed with a pinhole camera that is placed $40\ \text{m}$ away. If the camera is $20\ \text{cm}$ long, find the height of the image formed.
8. The length of a pinhole camera is $10\ \text{cm}$. It forms an image of linear magnification of 0.2. Find the position of the object.
9. State three characteristics of the image formed by a pinhole camera.
10. What can a shadow tell you about the size of the source of light?
11. Ammon is playing with a pinhole camera. He sees an image of a lamp on the screen of his camera.
    a. He moves the camera towards the lamp. What two things would he notice about the image?
    b. He holds the camera still and makes a very big hole in the front of the camera. The image is now very fuzzy What else would be different about the image? Explain your answer.
12. Why is it harder to see dark-coloured cars at night than light-coloured ones?
13. Choose words from this list to match each definition below.
    non-luminous inverted opaque penumbra umbra luminous upright translucent
    a. gives out light
    b. does not allow light to travel through it
    c. upside down (describing an image of an object)
    d. the very dark part of a shadow
    e. does not give out light
14. **(a)** Define the terms: angle of incidence, angle of reflection, and the normal ray.
    (b) What is the relationship between the angle of incidence and the angle of reflection?
15. Fig. 5.33 shows a plane mirror on which the angle of incidence is 30°.
    (a) What is the angle of reflection?
    (b) If the angle of incidence is increased to 40°, with the aid of a sketch diagram, show that the angle between the two reflected rays is 10°.

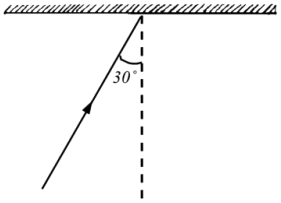


Fig. 5.33: An incident ray at an angle of 30°

16. The ray OA is incident on mirror M_1 as shown in Fig. 5.31. Draw a second plane mirror M_2 positioned such that the ray OA reflected by mirror M_1 is again reflected by the second mirror M_2 so as to reach the eye of the observer.

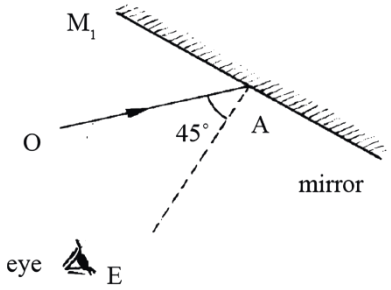


Fig. 5.31: An incident ray

17. **Figure 27.6** shows a ray of light PQ striking a mirror AB. The mirror AB and the mirror CD are at right angles to each other. QN is a normal to the mirror AB.

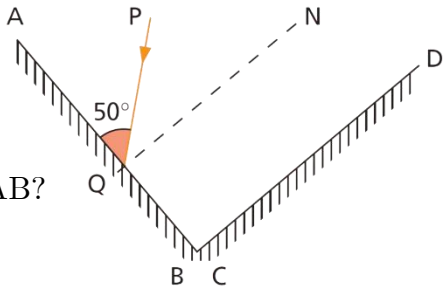


Figure 27.6

a What is the value of the angle of incidence of the ray PQ on the mirror AB?

b Copy the diagram, and continue the ray PQ to show the path it takes after reflection at both mirrors.

c What are the values of the angle of reflection at AB, the angle of incidence at CD and the angle of reflection at CD?

d What do you notice about the path of the ray PQ and the final reflected ray?

18. A ray of light strikes a plane mirror at an angle of incidence of $60°$, is reflected from the mirror and then strikes a second plane mirror placed so that the angle between the mirrors is $45°$. What is the angle of reflection at the second mirror, in degrees?

19. Explain why the reflection of light off ground glass changes from diffuse to specular if you spill water on it.

20. If the angle of incidence of a ray of light is $42°$, what is each of the following?

a. the angle of reflection

b. the angle the incident ray makes with the mirror

c. the angle between the incident ray and the reflected ray

21. If a light ray reflects off a plane mirror at an angle of $35°$ to the normal, what was the angle of incidence of the ray?

22. Light from a laser strikes a plane mirror at an angle of $38°$ to the normal. If the laser is moved so that the angle of incidence increases by $13°$, what is the new angle of reflection?

23. Two plane mirrors are positioned at right angles to one another. A ray of light strikes one mirror at an angle of $30°$ to the normal. It then reflects toward the second mirror. What is the angle of reflection of the light ray off the second mirror?

24. A ray of light strikes a mirror at an angle of $38°$ to the normal. What is the angle that the reflected angle makes with the normal?

25. A ray of light strikes a mirror at an angle of $53°$ to the normal.

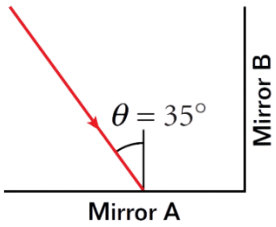
a. What is the angle of reflection?

b. What is the angle between the incident ray and the reflected ray?

26. A ray of light incident upon a mirror makes an angle of $36°$ with the mirror. What is the angle between the incident ray and the reflected ray?

27. A flat mirror that is $0.85\ \text{m}$ tall is attached to a wall so that its upper edge is $1.7\ \text{m}$ above the floor. Use the law of reflection and a ray diagram to determine if this mirror will show a person who is $1.7\ \text{m}$ tall his or her complete reflection.

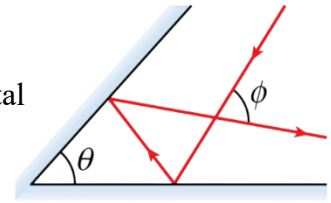
28. Two flat mirrors make an angle of $90.0°$ with each other, as diagrammed at right. An incoming ray makes an angle of $35°$ with the normal of mirror



a. Use the law of reflection to determine the angle of reflection from mirror

b. What is unusual about the incoming and reflected rays of light for this arrangement of mirrors?

29. The reflecting surfaces of two intersecting flat mirrors are at an angle of $\theta\ (0° < \theta < 90°)$, as shown in the figure below. A light ray strikes the horizontal mirror. Use the law of reflection to show that the emerging ray will intersect the incident ray at an angle of $\phi = 180° - 2\theta$.



30. Two adjacent plane mirrors form a right angle, as shown in **Figure 17-19.** A light ray is incident upon one of the mirrors at an angle of $30°$ to the normal.

a. What is the angle at which the light ray is reflected from the other mirror?

b. A retroreflector is a device that reflects incoming light rays back in a direction opposite to that of the incident rays. Draw a diagram showing the angle of incidence on the first mirror for which the mirror system acts as a retroreflector.

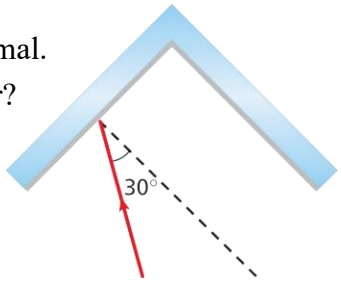


Figure 17-19